



Blood Donor Registry & Availability Search

A Smart Blood Donor Management System
with Machine Learning & Intelligent Search



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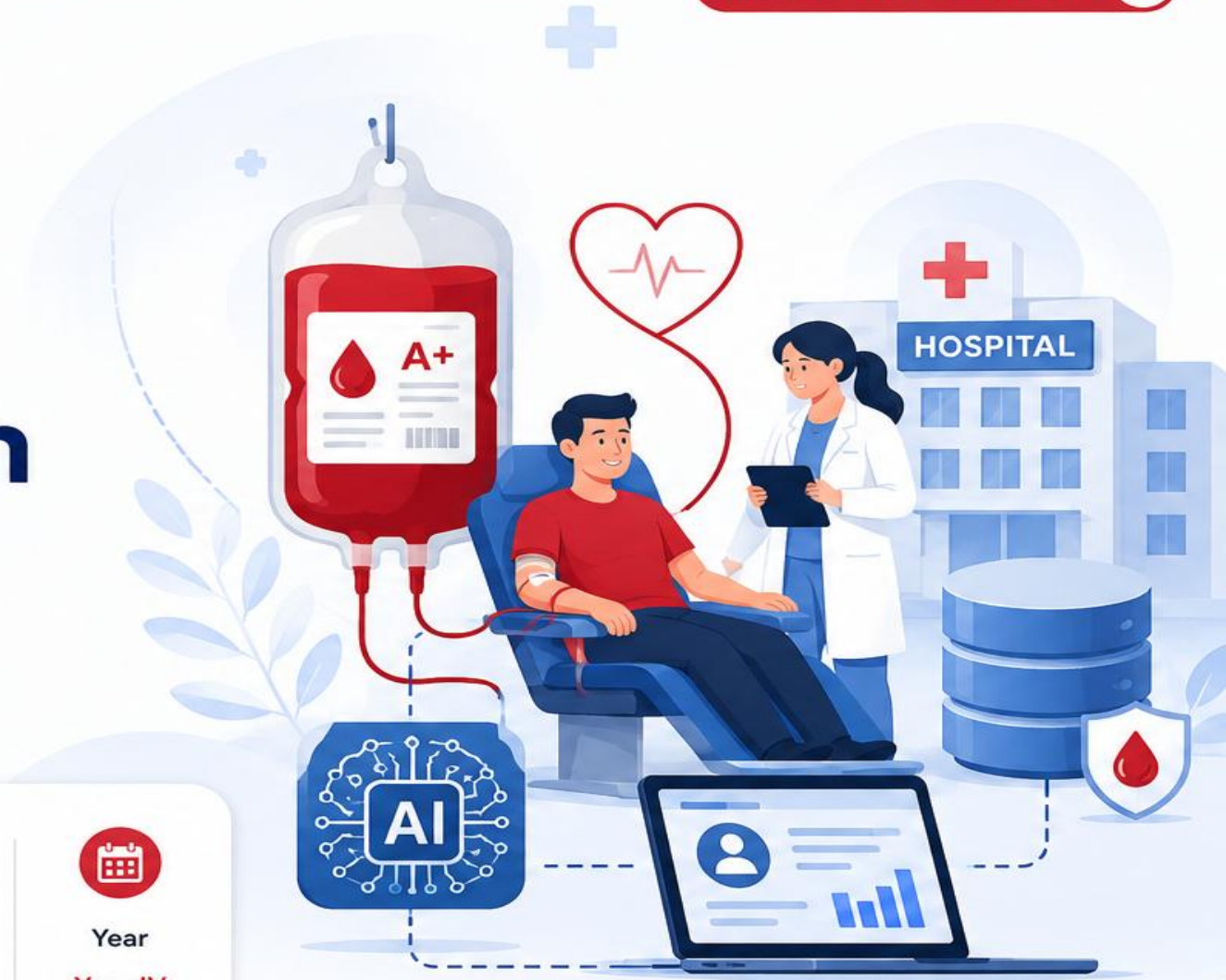
Department

Artificial Intelligence
& Data Science



Year

Year IV



Problem Statement



In emergency situations, hospitals face multiple **challenges** in finding the right blood donors quickly and efficiently.



Key Problems



Outdated / Manual Records

Donor information is stored manually and often becomes inaccurate or outdated.



No Intelligent Filtering

Searching eligible or available donors based on specific criteria is time-consuming.



Incomplete Donation History

Many donor records miss complete donation history and previous behaviour.



Slow Search for Rare Blood Groups

Finding rare blood groups or nearby donors takes longer during emergencies.



No Prediction of Donor Response

No system exists to predict the likelihood of a donor responding to a blood request.



Lack of Centralized Management

Hospitals lack a unified platform to manage donors, history, eligibility and search.



Impact

- Delayed blood availability
- Lower emergency response rate
- Increased risk for patients' lives



Goal

Build a smart, centralized and intelligent blood donor registry to save time, improve accuracy and ultimately **save lives**.

Project Overview



The **Blood Donor Registry & Availability Search** is an **AI-powered intelligent system** that helps hospitals quickly find **eligible and available blood donors**. It maintains donor history, predicts donor response using Machine Learning, calculates eligibility based on donation intervals, and provides a smart assistant for natural language queries.



Key Highlights



Centralized Donor Registry

Store and manage donor information in one place.



Advanced Search & Filters

Search donors by blood group, area, eligibility, availability and more.



Automatic Eligibility Calculation

Eligibility is calculated based on last donation and rules.



ML Based Donor Response Prediction

Predicts likelihood of donor response using Machine Learning.



Natural Language Assistant

Ask questions in simple language and get answers.



Secure & Privacy Focused

Strong data security measures ensure privacy and integrity.

Why It Matters



Faster Donor Search

Reduces time in finding eligible donors during emergencies.



Better Decision Making

Accurate predictions and eligibility checks improve donor selection.



Saves More Lives

Timely blood availability leads to better patient outcomes.



Scalable & Future Ready

AI-powered architecture designed to grow and adapt to future needs.



Right Donor. Right Time. Right Place.



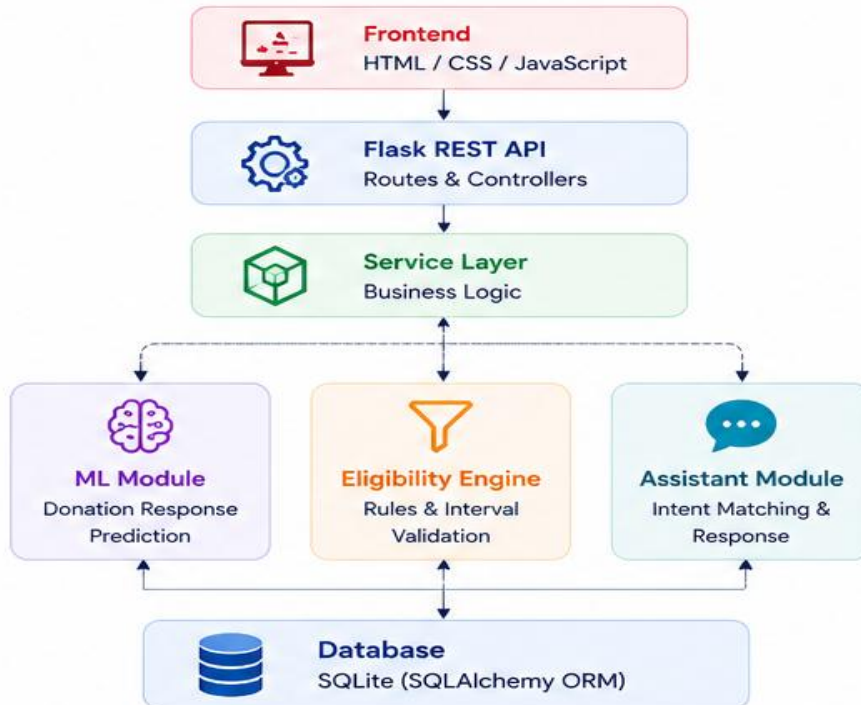
Technical Approach

We use a modern, modular and **AI-powered** approach to build a scalable Blood Donor Registry with **smart search**, eligibility engine and ML prediction.

Tech Stack

- Backend**
Python, Flask
- Database**
SQLite (SQLAlchemy ORM;
PostgreSQL-compatible schema)
- Machine Learning**
scikit-learn
(RandomForestClassifier)
- Assistant**
Rule-based
intent matching
- Frontend**
Plain HTML / CSS / JavaScript
(fetch-based, consumes REST API)
- Testing**
pytest

System Architecture



Key Components

- Smart Search & Filters**
Search donors by blood group, location, availability, age, gender and more.
- Eligibility Engine**
Validates donors based on last donation date, age, gender and health conditions.
- ML Prediction**
Predicts donor response probability using Random Forest Classifier.
- AI Assistant**
Answers user queries using rule-based intent matching.
- Data Management**
Stores and manages donor data securely with history tracking.
- Security & Validation**
Input validation, secure storage and role-based access.



How It Works



User Search
User searches for donors



Filters Applied
System applies eligibility and availability filters



ML Prediction
Donor response is predicted



Best Donor
Best matching donor is selected

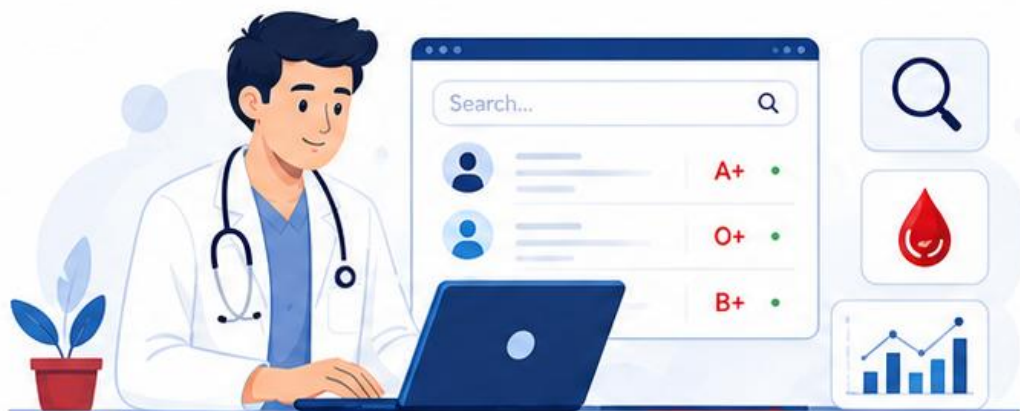


Contact Donor
Donor is contacted instantly

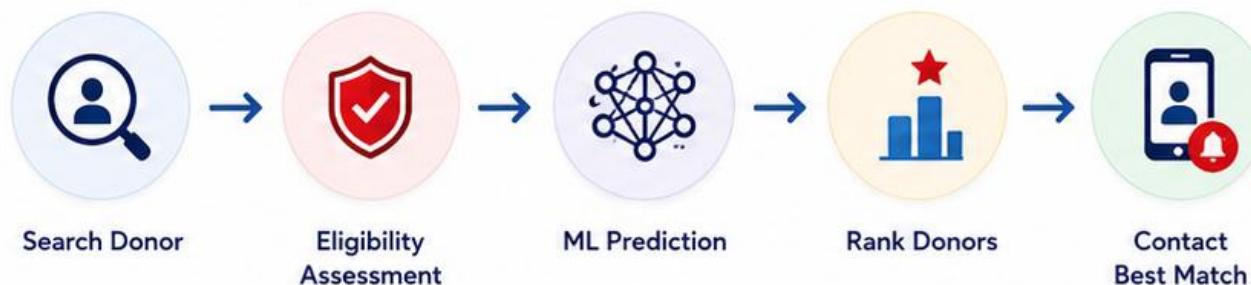
Intelligent Search & AI Decision Support



Combining intelligent eligibility assessment, machine learning prediction, and an intelligent assistant for faster donor identification.



Intelligent Decision Workflow



Advanced Search

Blood Group • Area • Availability



Intelligent Eligibility

Policy-Based Assessment • Health & Time Validation



ML Prediction

Response Probability • Confidence Level



Smart Assistant

Natural Language Queries • Instant Answers

Prediction Result



Likely to respond based on previous donation history and current availability.

Recommendation

✓ Eligible

✓ Available

✓ High Prediction Score



Best Matching Donor Selected

Novelty & Unique Features

Key innovations that make the Blood Donor Registry more intelligent, scalable, and assessment-ready.



1

Intelligent Eligibility Engine



Automatically computes donor eligibility using configurable age limits and a 90-day donation interval, with clear eligibility reasons and next eligible date.

2

AI Response Prediction



Random Forest model predicts donor response probability and confidence to help hospitals identify the most responsive donors.

3

Smart Donor Search



Advanced filtering by blood group, area, eligibility, and availability, with intelligent ranking of the most suitable donors.

4

Rule-Based Assistant



Natural language assistant that answers donor-related queries using existing business logic without duplicating rules.

5

Live Dashboard Analytics



Real-time KPI cards and interactive charts displaying donor statistics, eligibility distribution, blood group trends, and area-wise availability.

6

Data Integrity & Layered Architecture



Repository-Service architecture with SQL constraints, Pydantic validation, and modular design for reliability and maintainability.

Why Our Solution Stands Out



Centralized Registry



Rule-Based Eligibility



AI Prediction



Smart Search



Faster Emergency Response

Project Highlights



Repository-Service Layered Architecture



ML-Based Donor Response Prediction



Complete Donation History



Intelligent Search & Ranking



REST API-Based Integration



Assessment-Friendly Modular Design



AI Assists. Rules Validate. Humanity Saves Lives.



Impact & Future Scope

Creating a healthier community today and a smarter, more connected system for tomorrow.



Our Impact



Improved Access

Helps hospitals quickly find eligible donors and reduce critical delays.



Better Utilization

Ensures the right blood is available at the right time and right place.



Data-Driven Decisions

Machine Learning predictions improve donor outreach success.



Transparency & Trust

Secure, accurate and reliable system builds trust among donors and hospitals.

Key Benefits



For Hospitals

Faster donor search, better inventory management and improved patient care.



For Donors

Easy registration, track history and contribute to saving lives.



For Admins

Centralized control, data integrity and powerful analytics at a glance.



For Society

Stronger community network and better preparedness for emergencies.



For the System

Scalable, secure and intelligent platform for long-term sustainability.



For Partners

APIs enable seamless integration with other healthcare ecosystems.

Future Scope



Mobile Application

Dedicated mobile app for donors, hospitals and admins.



Advanced AI Models

Deep learning models for better prediction, personalized outreach and insights.



Interoperability

Integration with national blood banks and healthcare platforms.



Smart Notifications

Automated alerts for donations, eligibility reminders and urgent requests.



Global Expansion

Multi-language support and expansion to reach a global donor network.

Impact at a Glance



1,245+

Total Donors Registered



786+

Eligible Donors Today



652+

Available Donors Now



48+

Hospitals Connected



2,350+

Units of Blood Managed



91%

Prediction Model Accuracy



40%

Faster Donor Search



Countless

Lives Impacted

“ Technology connects us. **Humanity saves us.**
Together, we build a healthier and stronger tomorrow.



Thank You!

For your time, support and valuable feedback.



Together, we can build a smarter system
and **save more lives.**



“

One donation
can save three lives.

**One system can
save thousands.**



Donate Blood.
Save Lives.



Let's build a healthier,
stronger and smarter
tomorrow.

Technology connects us. Humanity saves us.